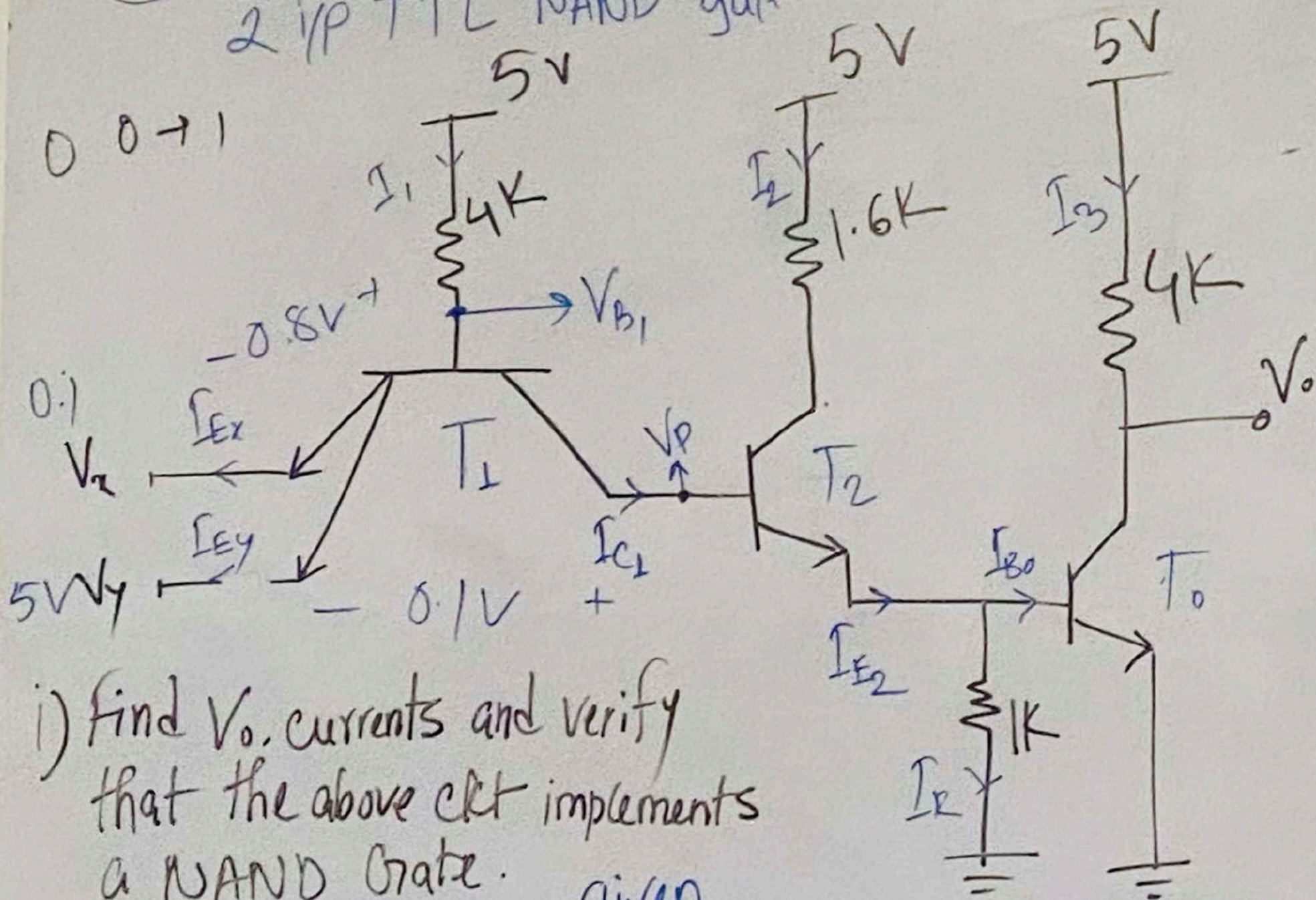


TTL (Transistor-Transistor logic)

2 i/p TTL NAND gate.



i) Find V_o , currents and verify that the above ckt implements a NAND Gate.

given

ii) Find P_T , P_{avg} . $V_{BE(sat)} = 0.8V$ | $V_{IL} = 0.1V$ | $\beta = 0.1$
 $V_{CE(sat)} = 0.1V$ | $V_{IH} = 5V$ | $R = 1K$

Case-1

$$V_x = 0.1V, V_y = 0.1V$$

E_1B, E_2B junction $\rightarrow$ FB
 CB junction has to be $\rightarrow$ FB

$T_1 \rightarrow$ Saturation

$$V_{BE} = 0.8V, V_{CE} = 0.1V$$

$$V_p = V_{CE} + V_x = 0.2V \rightarrow$$

$T_2, T_3 \rightarrow$ cutoff

$$I_{e1} = I_2 = I_{E2} = I_x = I_{B3} = I_3 = 0$$

$$I_3 = \frac{5 - V_o}{4K}$$

$$\Rightarrow 0 = 5 - V_o \rightarrow V_o = 5V$$

$$V_{B1} = 0.8 + 0.1 = 0.9V$$

$$I_1 = \frac{5 - 0.9}{4K} = 1.025mA$$

$$I_{Ex} = I_{Ey} = \frac{I_1}{2} = 0.512mA$$

$$P_1 = (5 - 0.1) \times (I_{Ex} + I_{Ey})$$

$$= (5 - 0.1) \times I_1 = 5.0225 \text{ mW}$$

Case-A

$$V_x = 0.1 \text{ V}, V_y = 5 \text{ V}$$

$E_1 B \rightarrow FB$, $E_2 B$ junc $\rightarrow RB$

$\therefore CB$ junction $\rightarrow FB$

$I_1 \rightarrow \text{saturation}$ $\therefore V_p = 0.2 \text{ V}$

$\therefore T_2, T_o \rightarrow \text{cutoff}$

$$I_{C1} = I_2 = I_{E2} = I_E = I_{B1} = I_3 = 0$$

$$V_0 = 5 \text{ V} \quad V_{B1} = 0.9 \text{ V}$$

$$I_1 = \frac{5 - 0.9}{4 \text{ k}} = 1.025$$

$$I_{Ey} = 0 \quad I_{Ex} = I_1$$

$$P_2 = (5 - 0.1) \times I_1 = 5.0225 \text{ mW}$$

EBJ

FB

FB

RB

RB

C-BJ

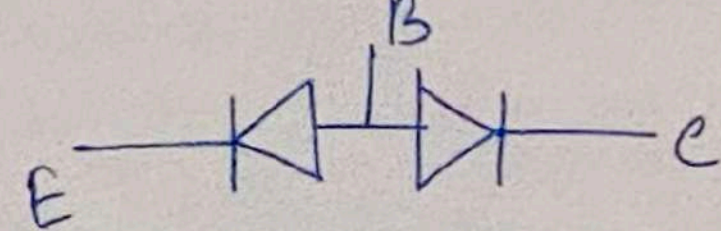
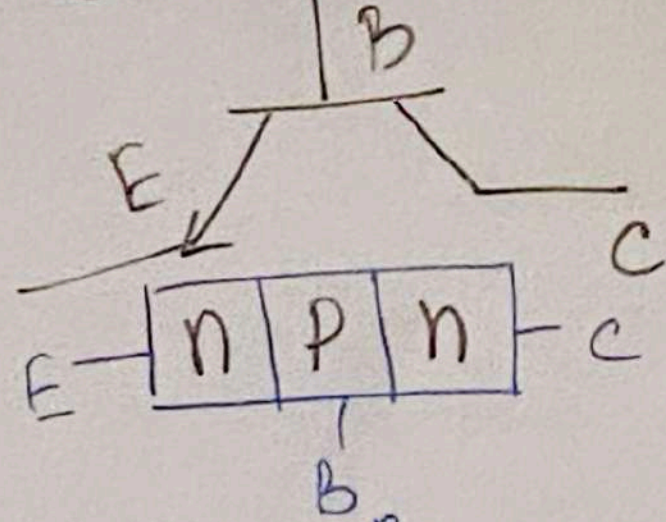
FB $\rightarrow$ Sat.

RB $\rightarrow$ FA

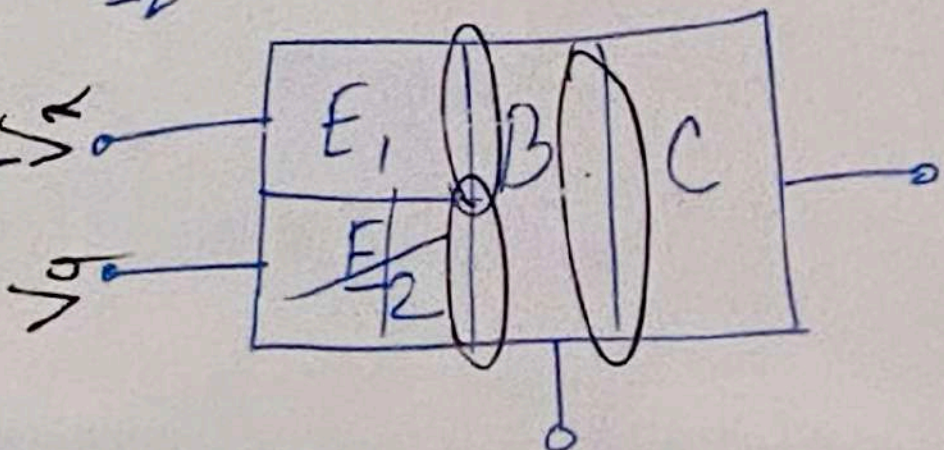
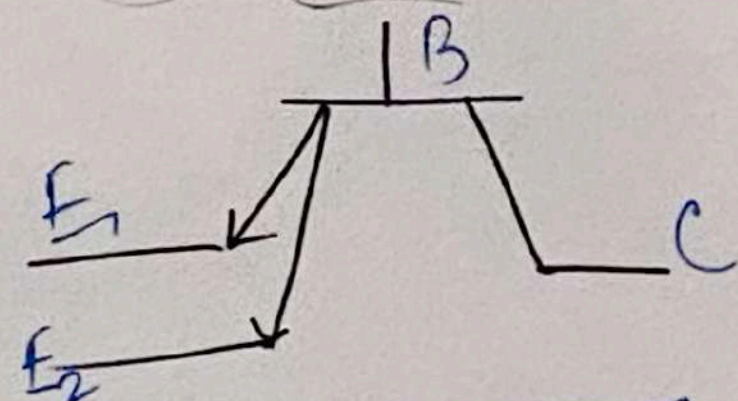
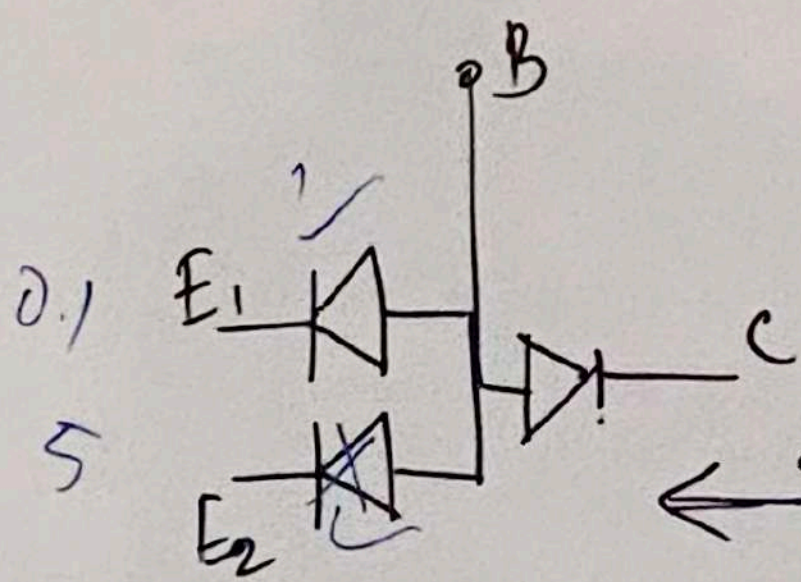
RB $\rightarrow$ cutoff

FB $\rightarrow$ RA

npn BJT

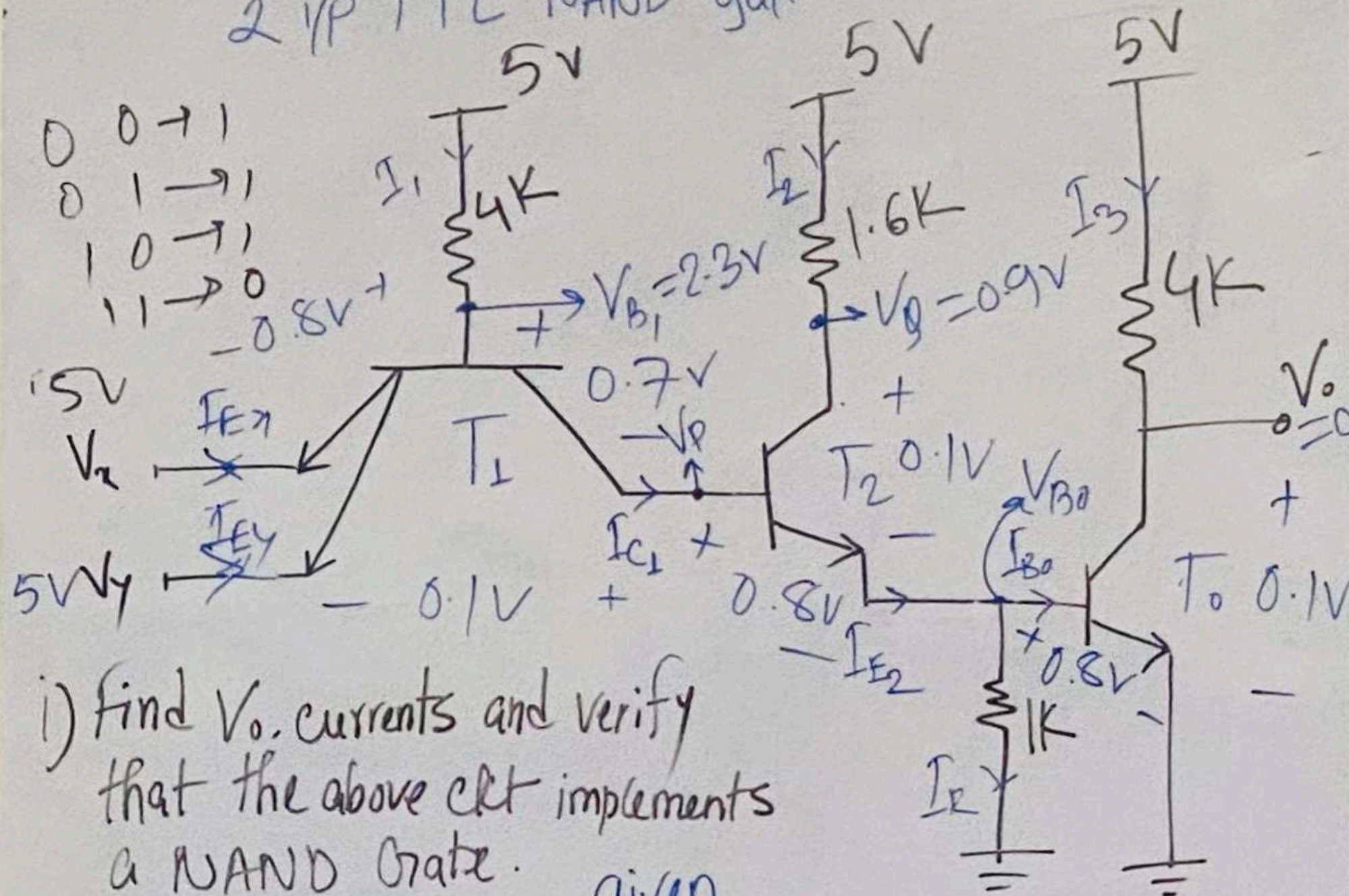


Multi-Emitter BJT



TTL (Transistor-Transistor logic)

2 i/p TTL NAND gate.



0	0	→	1
0	1	→	1
1	0	→	1
1	1	→	0

5V
V_g
5V
V_y

i) Find V_o , currents and verify that the above ckt implements a NAND Gate.

given

ii) Find P_T , P_{avg} .
 $V_{BE(sat)} = 0.8V$ | $V_{IL} = 0.1V$ | $\beta = 0.1$
 $V_{CE(sat)} = 0.1V$ | $V_{IH} = 5V$

Case-III

Same as case II

$$P_3 = P_2$$

Case-IV

$$V_x = 5V, V_y = 5V$$

E_B, E_2B junction → RB

CB junction → FB

T_1 → Reverse Active

T_2, T_3 → Sat. → $V_{BE} = 0.8V$
 $V_{CE} = 0.1V$

$$V_o = 0.1V \quad V_{Bo} = 0.8V$$

$$V_p = 0.8 + 0.8 = 1.6V$$

$$V_g = 0.8 + 0.1 = 0.9V$$

$$V_{B1} = 0.7 + 0.8 + 0.8 = 2.3V$$

$$I_1 = \frac{5 - 2.3}{4k} = 0.67 \text{ mA}$$

$$I_2 = \frac{5 - 0.9}{1.6k} = 2.56 \text{ mA}$$

$$I_3 = \frac{5 - 0.1}{4k} = 1.225$$

$$\beta_F = \frac{I_E}{I_B} = \frac{I_{E2}}{I_1}$$

$$\therefore I_{E2} = \beta_F \times I_1 = 0.1 \times 0.67 = 0.067 \text{ mA}$$

$$I_{E1} = \beta_F \times I_1 = 0.067 \text{ mA}$$

$$I_{E1} = I_1 + I_{E2} + I_{E1} = 0.804 \text{ mA}$$

$$I_{E2} = I_{E1} + I_2 = 3.364 \text{ mA}$$

$$I_F = \frac{0.8 - 0}{1} = 0.8 \text{ mA}$$

$$I_{B2} = I_{E2} - I_F = 2.564$$

$$P_4 = (5 - 0) \times (I_{E2} + I_1 + I_2 + I_3)$$

$$= 22.92 \text{ mW}$$

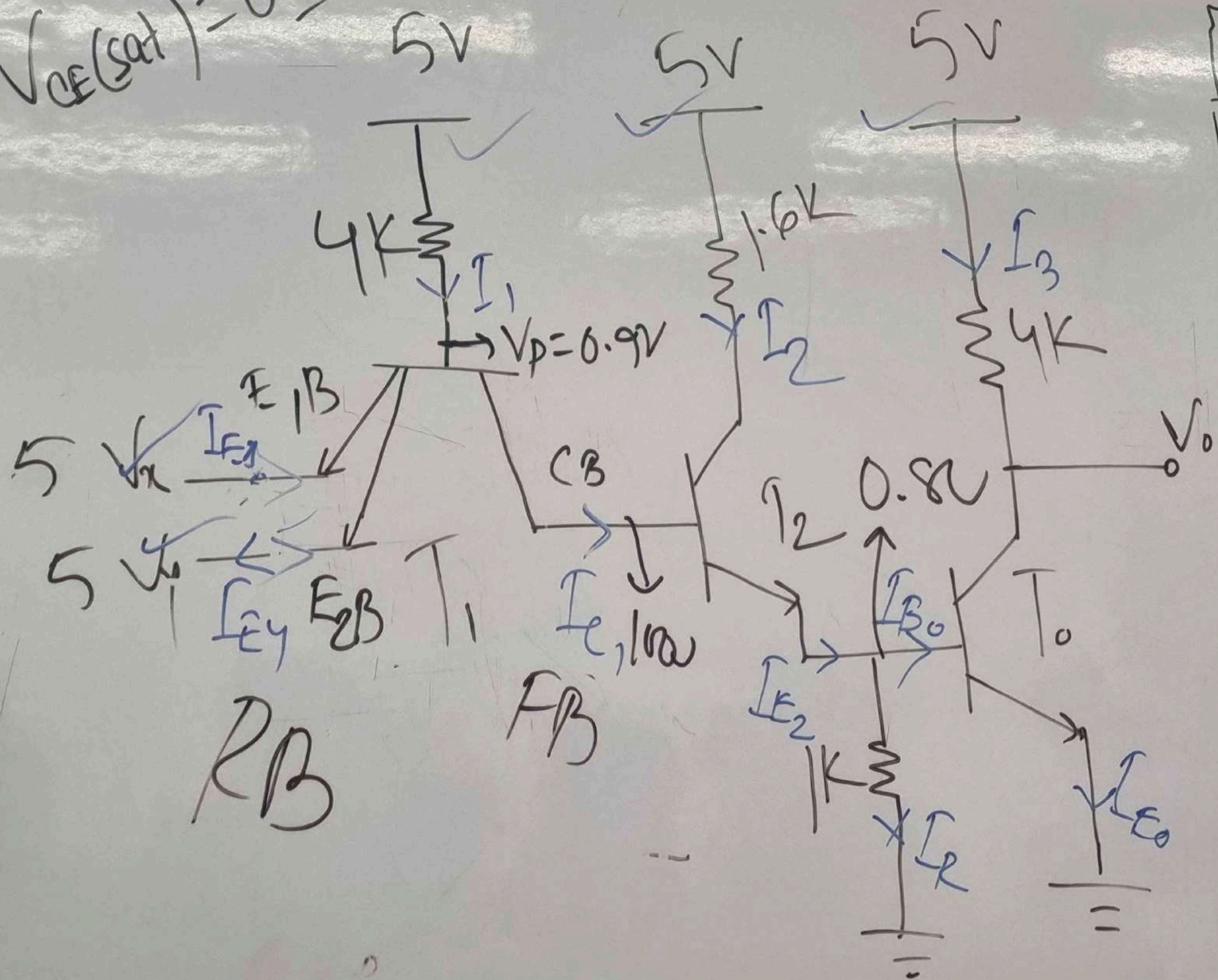
$$P_T = P_1 + P_2 + P_3 + P_4$$

$$P_{AVG} = \frac{P_T}{4}$$

3V

Power Consumption of TTL

$$V_{CE(sat)} = 0.1V$$



V_A	V_Y	V_O
0.1	0.1	5
0.1	5	5
5	0.1	5
5	5	0.1

NAND

Case-1
 $V_A = 0$

$I_1 =$

0.1V
p.

V_x	V_y	V_o
0.1	0.1	5
0.1	5	5
5	0.1	5
5	5	0.1

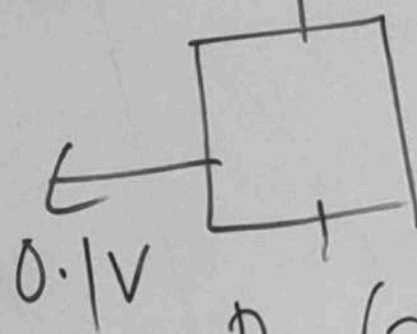
NAND

Case-I

$$V_x = 0.1V, V_y = 0.1V$$

$$I_1 = 1.025 \text{ mA}$$

$$V = 5V$$



$$P_1 = (5 - 0.1) \times I_1$$

$$= 4.9 \times 1.025$$

$$= 5.0225 \text{ mW}$$

Case-II

$$V_x = 0.1V, V_y = 5V$$

$$I_1 = \frac{5 - 0.9}{4}$$

$$= 1.025 \text{ mA}$$

$$P_2 = (5 - 0.1) I_1$$

$$= 5.0225 \text{ mW}$$

Case-III

$$V_x = 5V, V_y = 0.1V$$

$$\text{Similarly} \rightarrow P_3 = 5.025 \text{ mW}$$

Case-IV

$$V_x = 5V, V_y = 5V$$

$$I_{E1} = 0.0675$$

$$I_{E2} = 0.0675$$

$$I_1 = 0.675, I_2 = 2.56, I_3 = 1.23 \text{ mA}$$

$$P_4 = (5 - 0) (I_1 + I_2 + I_3)$$

$$= 23 \text{ mW}$$

$$\text{Total} \rightarrow P_T = P_1 + P_2 + P_3 + P_4 = 38.06 \text{ mW}$$

$$\text{Avg} \rightarrow P_{AVG} = \frac{P_T}{4} = 9.51 \text{ mW}$$

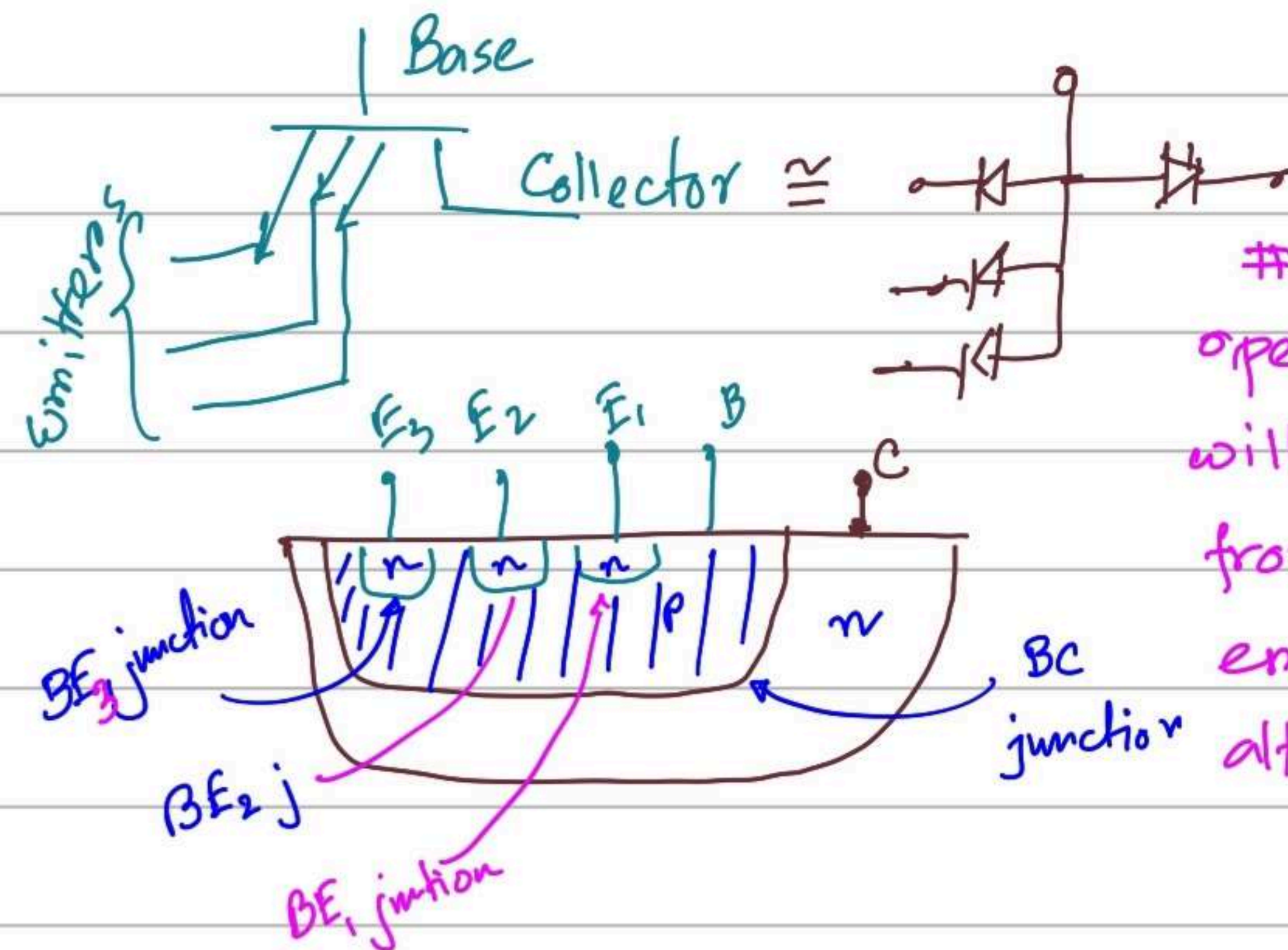
"Transistor-Transistor Logic"

Bipolar, $\rightarrow$ Saturated $\rightarrow$ TTL

BJT Switching time is faster than DTL.

The input terminals are connected to the emitter terminal of a transistor and the output terminal is connected to a switching transistor.

input terminals will be connected to multi-emitter transistor.



Transistor's operating mode will be determined from three emitter inputs altogether.

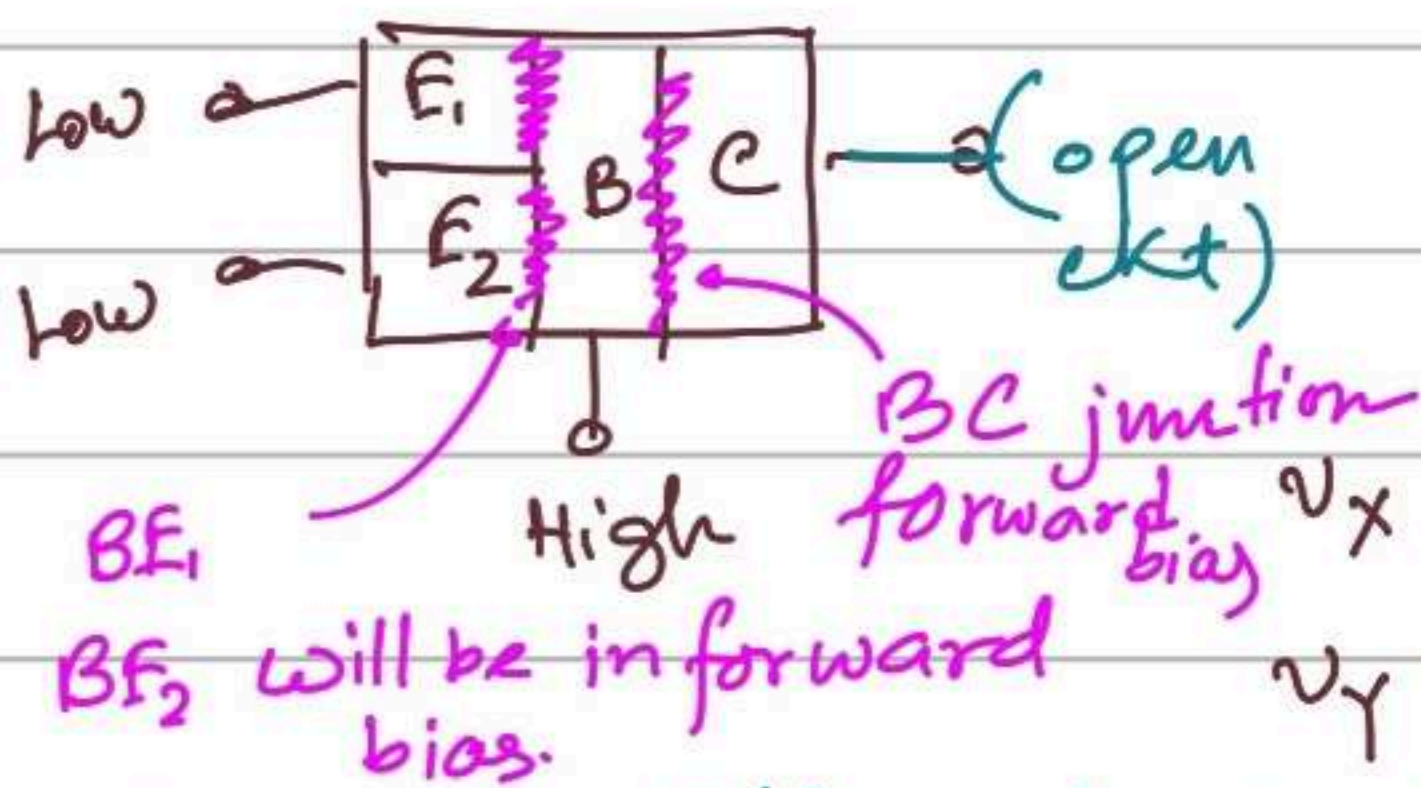
Basic Operation:

Find out the current and voltage of the following gate. Verify that it works like a NAND gate.

Case ①. $\underline{v_x = v_y = 0.1V}$

[Neaman] $v_{CE}(\text{sat}) = 0.1V$
 $v_{BE}(\text{sat}) = 0.8V$

For T_1 transistor



Our assumption is that T_2, T_0 cutoff.

According to our assumption

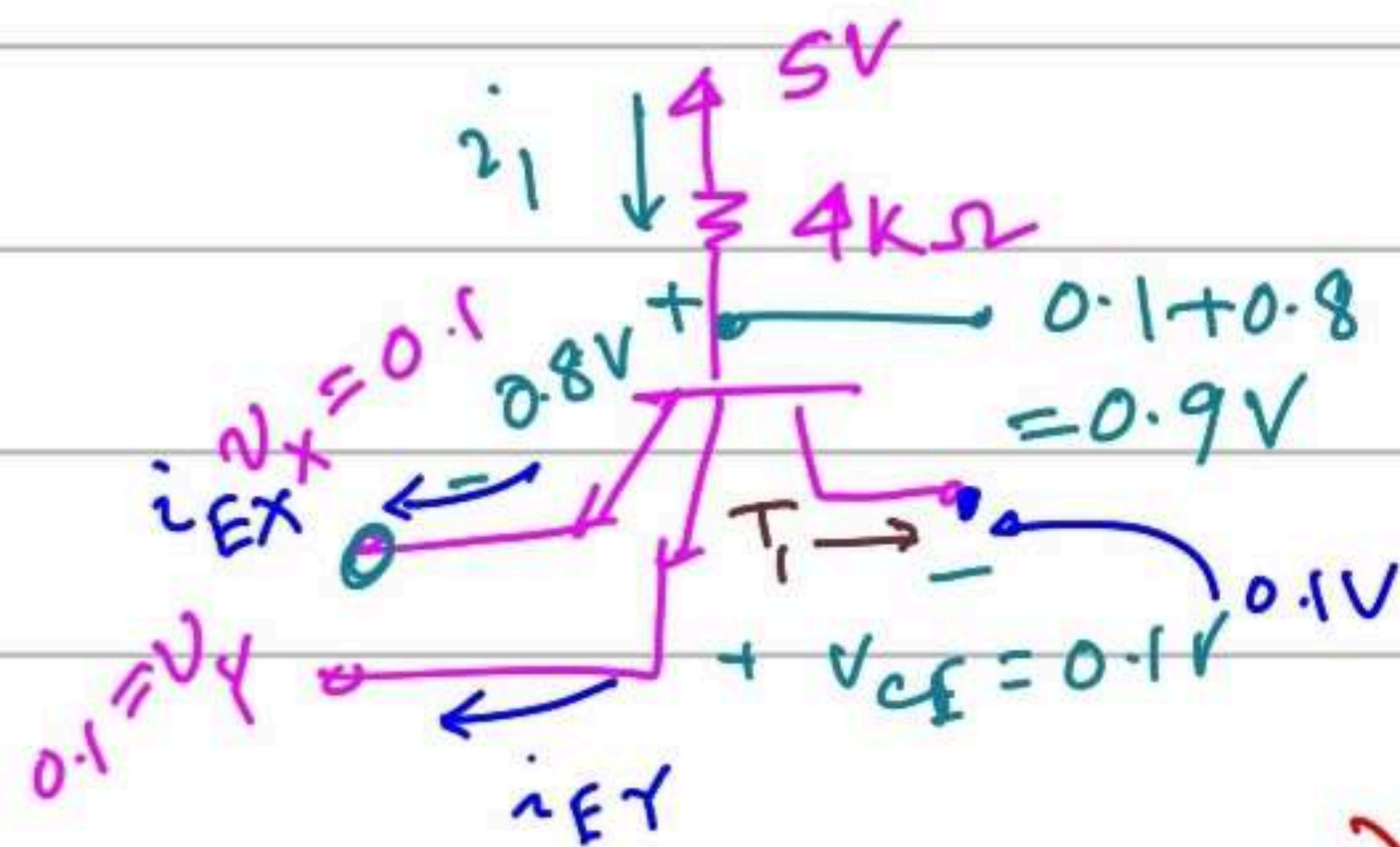
$$i_2 = i_3 = i_R = i_{B0} = i_{E2} = i_{C1} = 0 \text{ mA}$$

Bonus Question: B.C terminal's bias?

Ans: If B.C reverse bias, T_1 will be in forward active but for forward active we need, $i_C = \beta i_B$. But $i_C = 0$. Therefore not possible.

Hence T_1 must be in saturation mode. $\Rightarrow$

BC-junction is in forward bias

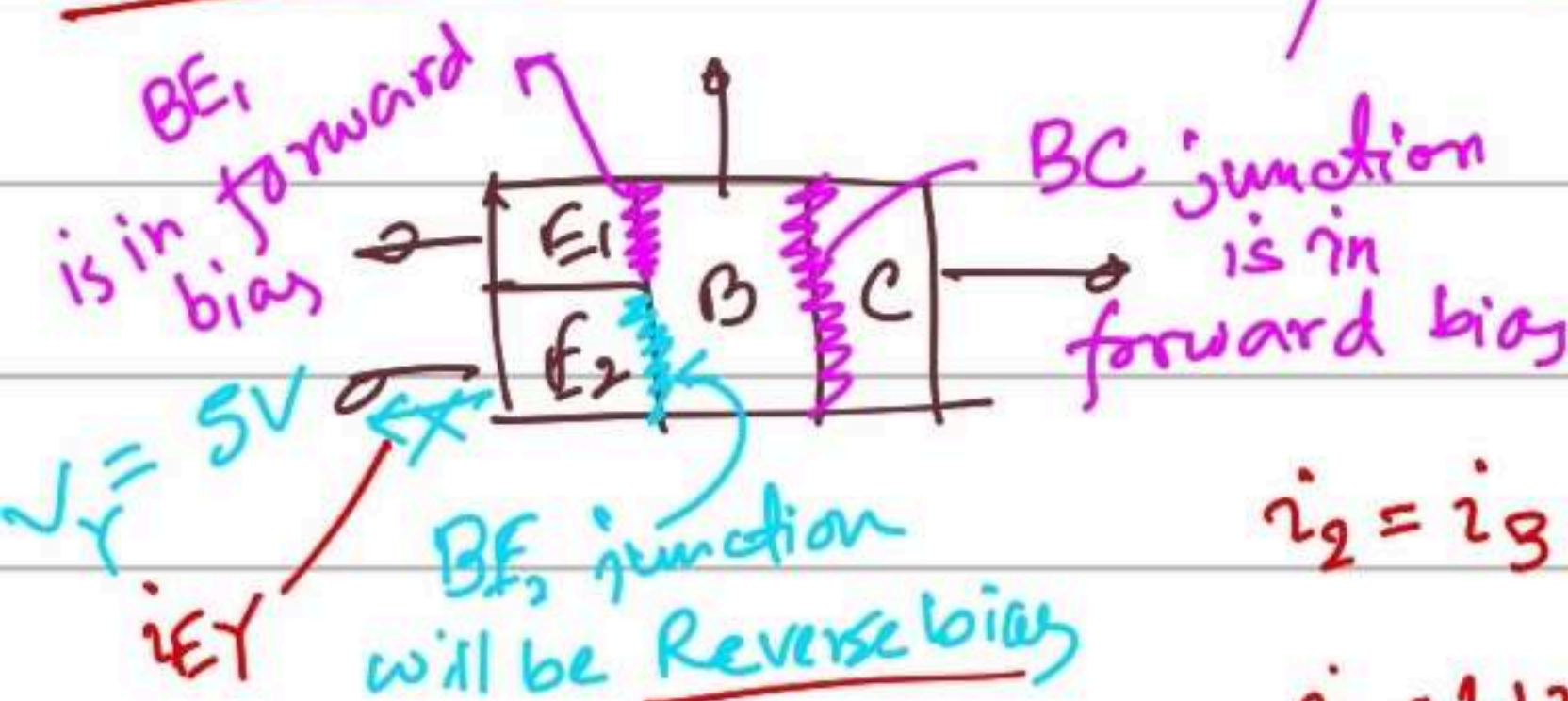


$$i_1 = \frac{5 - 0.9}{4 \text{ k}\Omega} = 1.125 \text{ mA}$$

$$i_{EX} = i_{EY} = \frac{i_1}{2} = 0.5675 \text{ mA}$$

$$\beta_{\text{forced}} = \frac{i_C}{i_B} = 0 < \beta_F \text{ Our assumption is valid.}$$

Case ② ($V_X = 0.2 \text{ V}, V_Y = 5 \text{ V}$) / $V_Y = 0.2 \text{ V}, V_X = 5 \text{ V}$



$T_1 \rightarrow$ saturation.

$T_0, T_2 \rightarrow$ cutoff.

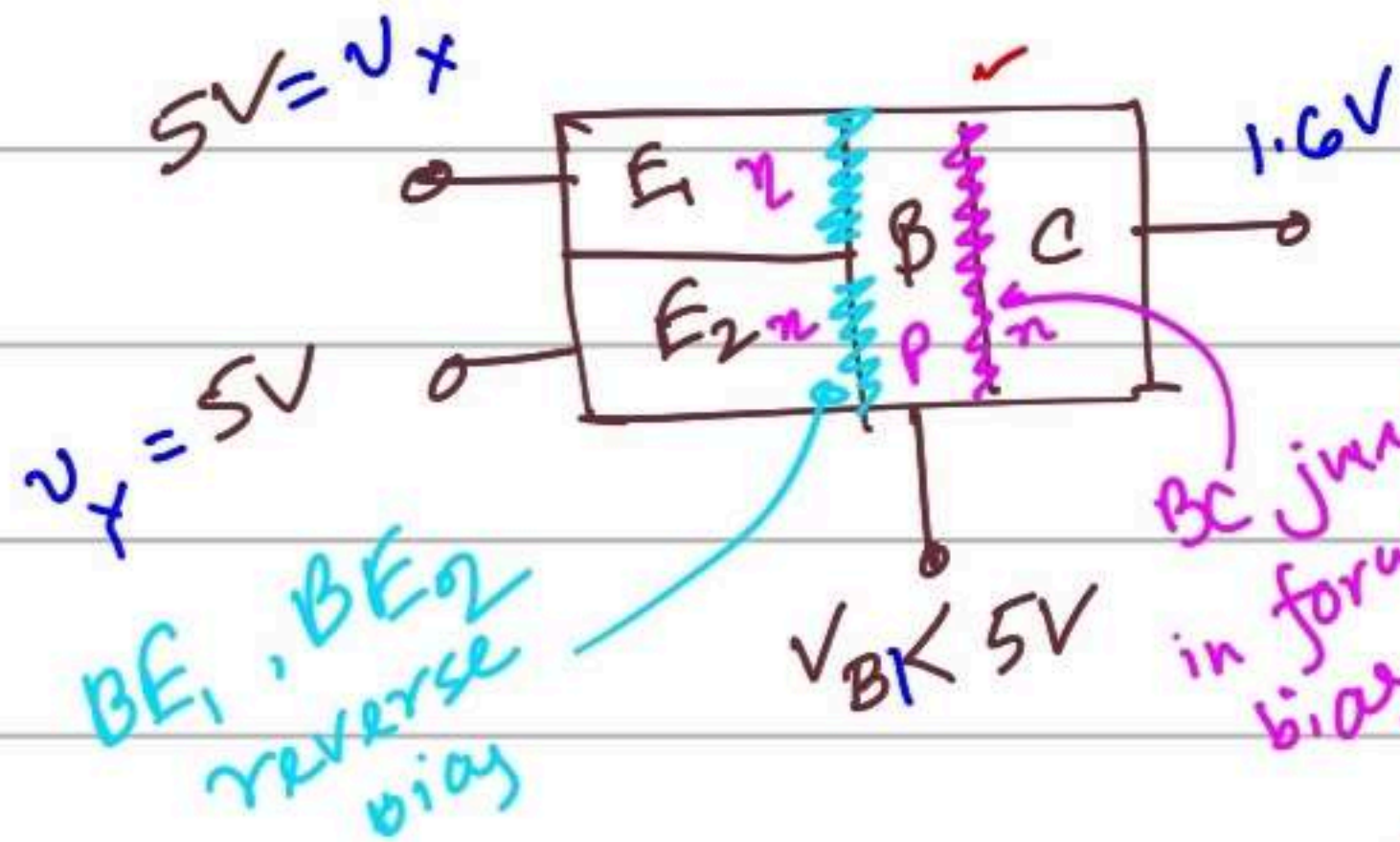
$$i_2 = i_3 = i_{B0} = i_{E2} = i_{C1} = i_R = 0$$

$$i_1 = 1.125 \text{ mA}, i_{EX} = i_1, i_{EY} = 0$$

Case 3: $v_x = v_y = 5V$ | Assumption:

$T_0, T_2 \rightarrow$ saturation

$$v_{CE}(\text{sat}) = 0.1V, v_{BE}(\text{sat}) = 0.8V$$



BC junction is in forward bias

$$\beta_F = 25$$

$$\beta_R = 0.1$$

T_1 transistor must be in reverse active mode.

Collector and Emitter terminals will switch their roles.

$v_{BC}(\text{reverse active}) = 0.7V$. [because BC junction is in forward bias]

$(i_{EX} = \beta_R i_B = i_{EY})$ β_R = reverse active current gain.

$$i_{B1} = i_1 = \frac{5 - 2.3}{4k} = 0.675mA, i_2 = \frac{5 - 0.9}{1.6k} = 2.5625mA$$

$$i_3 = \frac{5 - 0.1}{4k} = 1.225mA, i_{EX} = i_{EY} = \beta_R i_{B1} = 0.0675mA$$

$$i_{C1} = i_{EX} + i_{EY} + i_{B1} = 0.81mA, i_{E2} = i_2 + i_{C1} = 3.3725mA$$

$$i_R = \frac{0.8 - 0}{1k} = 0.8mA, i_{B0} = i_{E2} - i_R = 2.5725mA$$

$$\beta_{\text{forced}}(T_2) = \frac{i_2}{i_{C1}} = \frac{2.5625}{0.81} = 3.16 < \beta_F = 25$$

$$\beta_{\text{forced}}(T_0) = \frac{i_3}{i_{B0}} = \frac{1.225}{2.5725} \approx 0.47 < \beta_F = 25$$

Our assumptions are valid.

v_x	v_y	v_o
0.1	0.1	5
0.1	5	5
5	0.1	5
5	5	0.1

This circuit works like a NAND gate.